

Study of Autonomous Vacuum Robot Navigation System and Comparison with UAV Path Planning

Details:

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Introduction

I studied the working of autonomous robotic vacuum cleaners and the navigation techniques used in these systems. Unlike normal vacuum cleaners, these robots can move automatically, detect obstacles, and clean the area with **minimum human** assistance.

The autonomous robotic vacuum cleaner uses different sensors to collect information about its surroundings. The sensor data helps the robot detect obstacles such as walls and furniture, estimate its position inside the room, and decide a suitable path for movement.

During this study, I understood that the concepts used in robotic vacuum cleaners are also important for other **autonomous systems** such as drones and mobile robots. Sensors, mapping, localization, and path planning are important techniques that help these systems operate safely without continuous human control.

The objective of this study is to understand how an autonomous robot senses its environment, creates a map, plans its movement, and avoids obstacles.

Robot Architecture and Working Principle

During my study, I understood that an autonomous vacuum cleaner works by integrating sensing, processing, navigation, and cleaning systems.

The robot first collects information about its surroundings using sensors such as LiDAR, camera, IR sensors, and bumper sensors. This data helps the robot detect obstacles like walls and furniture.

The collected sensor information is processed by the controller, which helps in **localization and mapping** of the cleaning area. Based on the generated map, the path planning algorithm selects a suitable path for movement.

The motor controller controls the wheel movement according to the planned path, allowing the robot to move safely while avoiding obstacles.

During operation, the cleaning system performs the cleaning task using brushes, suction, and mopping mechanisms.

The overall working flow of an autonomous vacuum cleaner is:

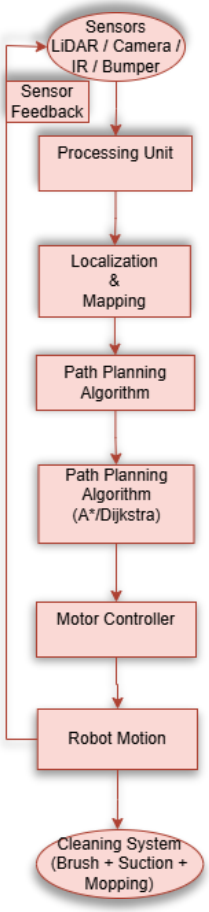


Figure: 1. Functional Architecture of Autonomous Robotic Vacuum Cleaner

Sensors Used in Autonomous Vacuum Cleaner

During my study of robotic vacuum cleaners, I understood that sensors are one of the most important parts of the robot. They help the robot to detect its **surroundings and move** safely. Different sensors are used for tasks such as obstacle detection, distance measurement, position estimation, and movement control.

Navigation System (Localization and Mapping)

Sensor's	Purpose
LiDAR Sensor	Measures distance from surrounding objects using laser pulses. It helps in creating a 2D floor map and detecting obstacles.
RGB / Depth Camera	Captures images of the surroundings. RGB cameras help in object detection, while depth cameras provide distance information.
IR Sensor	Used for short-range obstacle detection and edge detection to prevent collisions or falling.

Bumper Sensor	Detects physical contact with obstacles and helps the robot change direction.
Wheel Encoder	Measures wheel rotation to estimate distance travelled and robot movement.
IMU Sensor	Measures robot orientation and motion using acceleration and rotation data.

Navigation System

During my study, I understood that the navigation system helps an autonomous vacuum robot to understand its position and move safely inside the cleaning area.

For autonomous movement, the robot mainly needs to answer two questions:

• Where am I? → Localization
• What does my surroundings look like? → Mapping

The robot uses **sensor data** from LiDAR, camera, wheel encoder, and IMU to estimate its position and understand the room environment. This information helps the robot decide its movement and avoid obstacles during cleaning.

Localization

Localization means finding the current position and direction of the robot inside the room, The robot continuously updates its position using sensor information. It helps the robot to know:

- Current location of the robot
- Movement direction of the robot
- Orientation of the robot

This allows the robot to move correctly and continue cleaning without losing its path, in the indoor.

Mapping

Mapping is the process by which the robot creates an understanding of its surrounding area.

In an autonomous vacuum cleaner, LiDAR and other sensor data are used to create a **2D floor map** of the room. This map helps the robot identify:

- Walls and room boundaries
- Obstacles like furniture

- Free areas for movement
- Cleaning coverage area

The generated map is used by the robot to decide its movement path and perform efficient cleaning.

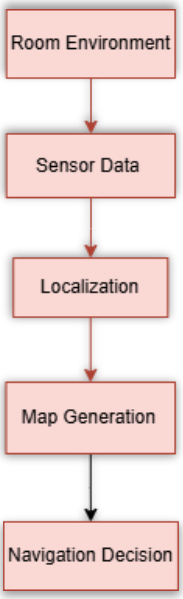


Figure: 2. Basic Navigation and Mapping Flow of Autonomous Vacuum Robot

SLAM (Simultaneous Localization and Mapping)

SLAM is a technique used in autonomous robots where the robot creates a map of an unknown environment while estimating its own position at the same time.

Working of SLAM:

The robot collects information from sensors such as LiDAR, camera, wheel encoder, and IMU. The sensor data is used to detect surrounding objects and calculate the robot's movement.

The robot then compares the new sensor information with the previous map and updates its position and map continuously.

The basic process is:

Sensors → Environment Data → Position Estimation → Map Update → Navigation Decision

SLAM helps the vacuum robot to understand the **room layout and move efficiently** even when a **predefined map** is not available.

Path Planning

After the robot creates the map of the room, the next task is to decide how it should move. During my study, I understood that path planning helps the robot **select a safe and efficient route** for cleaning.

The robot uses the generated map and sensor information to decide:

- Which area needs to be covered
- Which direction it should move
- How to avoid obstacles during movement

The main aim is to clean the maximum area with minimum repeated movement and less battery usage, For path planning, algorithms are used to calculate the movement path.

A* Algorithm

This A* algorithm is mostly used for finding an efficient (shortest path) path between the current position to the target position, It calculates the path by considering:

- Distance already travelled by the robot
- Estimated distance towards the target

Since the robot already has a map**, A*** can generate a suitable path while avoiding obstacles.

Dijkstra Algorithm

Dijkstra algorithm checks different possible routes and selects the shortest one, It can find an optimal path, but it may require more computation because it explores more possible paths.

Obstacle Avoidance

Obstacle avoidance is an important part of autonomous navigation that helps the vacuum robot move safely without collision, During movement, the robot continuously collects information from sensors such as LiDAR, camera, IR sensors, and bumper sensors to detect nearby objects like walls, furniture, and other obstacles.

When an obstacle is detected, the robot performs the following process:

1. Sensors detect the obstacle
2. Robot estimates its position relative to the obstacle
3. Navigation system updates the planned path
4. Motor controller changes the movement direction

This allows the robot to continue **cleaning while avoiding** collisions.

Comparison Between Robotic Vacuum Cleaner and Drone Navigation

During my study, I found that robotic vacuum cleaners and drones use similar autonomous navigation concepts, but their working environment and movement are different.

Aspect	Autonomous Robotic Vacuum Cleaner	Autonomous Drone / UAV
Working Environment	Works on indoor floor surfaces	Works in air with 3D surroundings
Movement	Uses wheels for movement	Uses propellers and motors for flight
Sensors	LiDAR, RGB Camera, IR Sensor, Bumper Sensor, Wheel Encoder, IMU	Camera, LiDAR, GPS, IMU, Ultrasonic Sensor
Position Estimation	Uses wheel movement, IMU and map data	Uses GPS, IMU and visual information
Mapping	Mainly creates a 2D room map	Uses 2D/3D mapping based on application
Navigation	Uses SLAM to understand the area and plan movement	Uses SLAM, GPS and sensor fusion for navigation
Path Planning	Finds a cleaning path to cover the floor area	Finds a safe flight path to reach the target
Obstacle Avoidance	Avoids walls, furniture and indoor objects	Avoids buildings, trees and other flying obstacles
Control System	Controls wheel motors	Controls multiple motors for stable flight

From this comparison, I understood that the basic idea is similar in both systems: sensors collect information, the controller processes it, and the robot/drone decides its movement. The main difference is that a vacuum robot works mostly in a **2D environment**, while a drone has to handle movement in **3D space**.

Proposed Sensor-Based Autonomous Navigation Prototype Idea

While studying autonomous robotic vacuum cleaners and UAV navigation, I understood that most of these systems basically work in a similar way. They first sense the environment, then process that information, and finally take movement decisions.

So, to practically understand this, I am thinking of making a small robot prototype which can move on its own, detect obstacles, and avoid them. The idea is not to build a full vacuum cleaner or advanced drone system, but just to understand how navigation actually works in real robots.

Basic Idea of How It Will Work

The basic flow of the system will be like this:

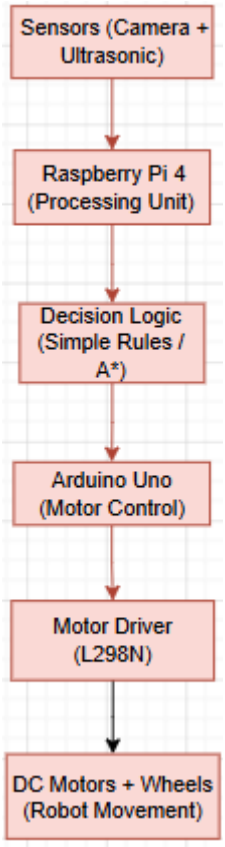


Figure: 3. Block Diagram of Proposed Raspberry Pi Based Autonomous Navigation System

So basically, the robot will keep checking its surroundings and decide where to go.

Hardware Setup

1. **Main Controller -**

   I am using Raspberry Pi 4 as the main processing unit of the system. **It is responsible** for collecting sensor data, processing camera input, running the Python program, and making navigation decisions based on the environment.

   The movement commands generated by Raspberry Pi are sent to the Arduino Uno, which controls the motors through the motor driver.

2. **Motor Control (Arduino Uno + L298N Motor Driver)**

I am using **Arduino Uno** for controlling the motor movement. The commands from Raspberry Pi will be given to Arduino, which will generate control signals for the motor driver. The L298N motor driver will provide the required current to drive the DC motors.

3. Robot Movement

I am planning to use a simple 2-wheel robot with a caster wheel for support. The robot will use DC motors connected with the L298N motor driver for forward, backward, and turning movement.

4. Power Supply

I will be using two separate power sources in the system. A **battery will** be used to run the motors, while a **power bank** will supply power to the Raspberry Pi. This separation is necessary because the motors require higher power, whereas the Raspberry Pi needs a stable 5V supply to function properly without interruptions.

5. Sensors – (Ultrasonic Sensor)

I will use ultrasonic sensor is used to check distance of obstacles in front of the robot. It keeps measuring how far objects are. If anything comes too close, the robot will stop or change its direction to **avoid hitting** it.

Pi Camera

I am using Pi Camera is used to capture what is in front of the robot. I will use simple **OpenCV image processing** to check whether something is **blocking the path or not**.

Software Understanding

The software will run in Python on Raspberry Pi.

1. Sensor Reading

The robot will continuously read the distance from the ultrasonic sensor and also capture images from the camera to understand the **environment** in front of it.

2. Camera Processing (OpenCV idea)

In this part, I will use **OpenCV** to process the **images coming** from the Pi Camera. The idea is to convert the image into a simple form and check if there is any object blocking the **path in front of the robot**.

Steps:

- Take image from the camera

- Convert it into grayscale
- Apply edge detection
- Check the center part of the image

If the center part looks blocked, then it is considered that an obstacle is in front of the robot.

3. Movement Logic

The robot will follow **simple rules** for movement. If an obstacle is detected in front, it will stop first, then move a little backward and check both left and right sides. If the left side is clear, it will turn left. Otherwise, it will turn right. If no obstacle is detected, the robot will simply move forward.

This is basically how obstacle avoidance works in the system.

4. Path Planning Idea (A*)

I also studied the A* algorithm. For now, I am keeping it only as a **basic concept**. The idea is that a room can be treated **like a grid**, where obstacles are marked in different cells. Then A* can be used to find the **shortest path** from the starting point to the goal.

This is mainly to connect with how real robotic vacuum cleaners plan their movement in a structured environment.

Conclusion

From this study, I understood that robotic vacuum cleaners mainly work by sensing the environment and then taking decisions based on that **information**. Basically, the robot detects obstacles, understands its position, and moves safely without any human control.

I also understood that the same basic concept is used in drones and other autonomous systems, but the main difference is the type of movement and environment. Ground robots work in **2D space**, while drones work in **3D space**, so the level of complexity changes.

While going through this, I was able to connect how sensor data is used in real time for simple **decision making and navigation**. It also made me realize that even a small system using Raspberry Pi and basic sensors can demonstrate the same idea in a practical way.

Overall, this study helped me understand that autonomous navigation is not one big complex system, but actually a combination of small simple steps working together.